F_1 male with double recessive female.

i) What type of linkage is seen?

ii) Draw the cross with correct genotype.

iii) What is the possible genotype in F_2 generation?

15.

S. no	Gamete types	Number of progenies
1.	ABC	349
2.	Abc	114
3.	abC	124
4.	AbC	5
5.	aBc	4
6.	aBC	116
7.	ABc	128
8.	abc	360

i) What is the name of this test cross?

ii) How will you construct gene mapping from the above given data?

iii) Find out the correct order of genes.

16. What is the difference between missense and nonsense mutation?

17. 

From the above figure identify the type of mutation and explain it.

18. Write the salient features of Sutton and Boveri concept.

19. Explain the mechanism of crossing over.

20. Write the steps involved in molecular mechanism of DNA recombination with diagram.

21. How is ***Nicotiana*** exhibit self-incompatibility. Explain its mechanism.

22. How sex is determined in monoecious plants. write their genes involved in it.

23. What is gene mapping? Write its uses.

24. Draw the diagram of different types of aneuploidy.

25. Mention the name of man-made cereal. How it is formed?

Glossary

Branch Migration: The process in which base pairs on homologous strands are consequently exchanged at a Holliday junction, moving the branch up or down the DNA sequence.

Cis configuration: The presence of dominant alleles of two or more pairs on one chromosome and the recessive alleles on the homologous chromosome.

Feminizing Masculinizing: To induce female characteristics in male To induce male characteristics in female

Heteroduplex: A double stranded molecule of nucleic acid originated through genetic recombination from different sources

Self incompatibility: A genetic mechanism which prevent self fertilization thus encourage outcross.

Synapsis: The pairing of two homologous chromosomes that occurs during meiosis.

Tassel seed: Feminization of the tassel

Trans configuration: An arrangement in which the dominant allele of one pair of genes and the recessive allele of another pair are on the same chromosome

Transesterification: A reaction that breaks and makes chemical bonds in a coordinated transfer, so that no energy is required.

Vestigial: Rudimentary organ of body become functionless in the course of evolution

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_053-077 CH 03.indd 67 3/14/2020 10:32:17 AM